

# 10.5.8. Basic Program Nodes: Waypoints

## Description

Waypoints are one of the most central parts of a robot program, telling the robot arm where to go one movement at a time.

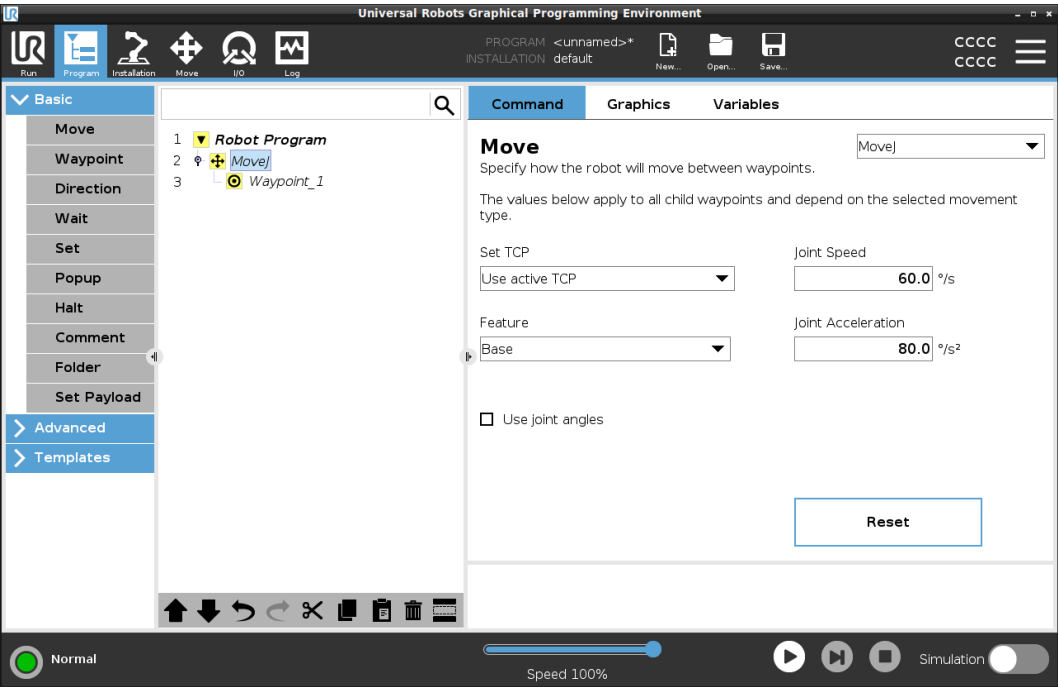
## Add Waypoints

A waypoint accompanies a Move, so adding a Move is required for the first waypoint.

## Add a waypoint to a robot program

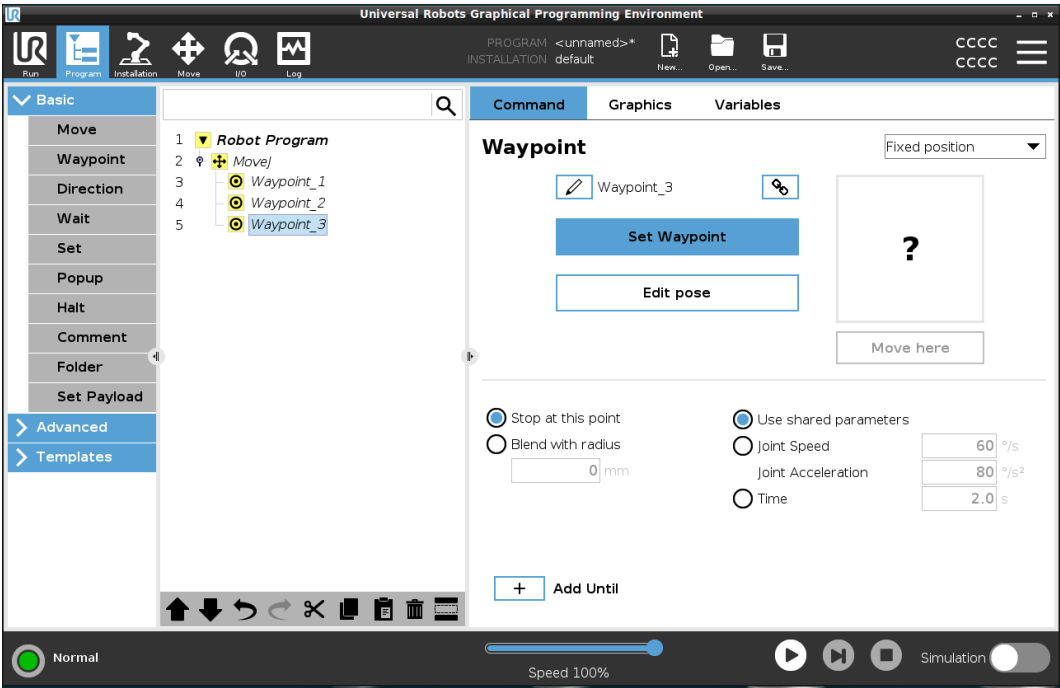
1. In your Robot Program, select the place where you wish to add a Move.
2. Under Basic, tap **Move**.

A waypoint is added to the robot program together with the Move node.



Add additional waypoints to a Move or Waypoint

1. In your Robot Program, select a Move node or Waypoint node.
2. Under Basic, tap **Waypoint**.  
The additional waypoint is added in the Move node. This waypoint is part of the Move command.



The additional waypoint is added under the waypoint that you selected in the robot program.

Detail

Using a waypoint means applying the taught relationship between the feature and the TCP from the Move command. The relationship between the feature and the TCP, applied to the current selected feature, achieves the desired TCP location. The robot calculates how to position the arm to allow the current active TCP to reach the desired TCP position.

10.5.9. Set Payload

Description

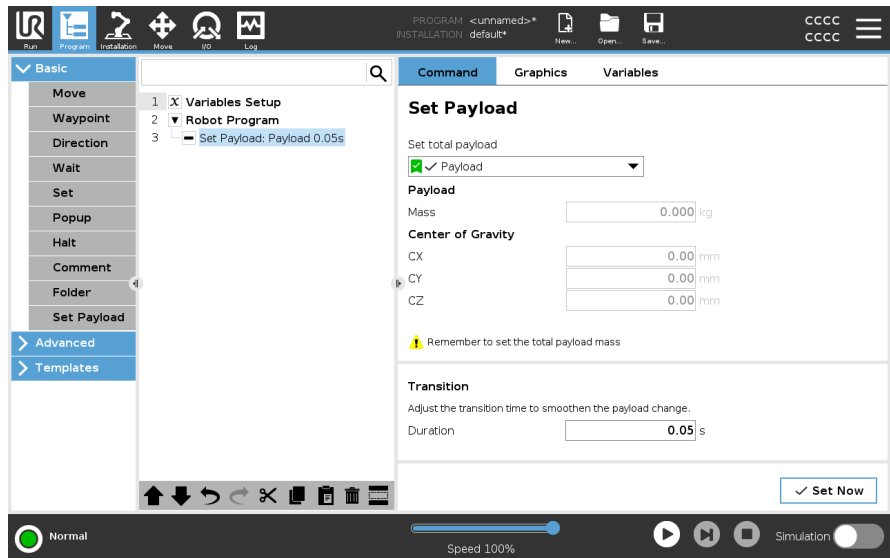
The Set Payload command allows you to configure the payload for the robot. Payload is the combined weight of everything attached to the robot tool flange.  
When to use:

- When adjusting the payload weight to prevent the robot from triggering a robot stop. A correctly configured payload weight ensures optimal robot movement. Setting the payload correctly ensures optimal motion performance and avoids robot stops.
- When setting up the payload for use in a pick and place program, using a gripper.

## Set Payload

### Use the Set Payload command

1. In your robot program, select the place or node where you wish to add a Set command.
2. Under Basic, tap **Set Payload**.
3. Use the drop-down, under **Select Payload**.
  - a. Select one of the payloads already configured.
  - b. Or, use the drop-down to configure a new payload by selecting **Custom Payload** and completing the mass and CoG fields.



**Tip** You can also use the **Set Now** button to set the values on the node as the active payload.

**Use tip** Remember to always update your payload when making any changes to the configuration of the robot program.

**Example: Set Payload** In a pick and place program, you would create a default payload in the installation. Then you add a Set Payload when picking up an object. You would update the payload after the gripper closes, but before starting to move. Additionally, you would use the Set Payload after the object has been released.

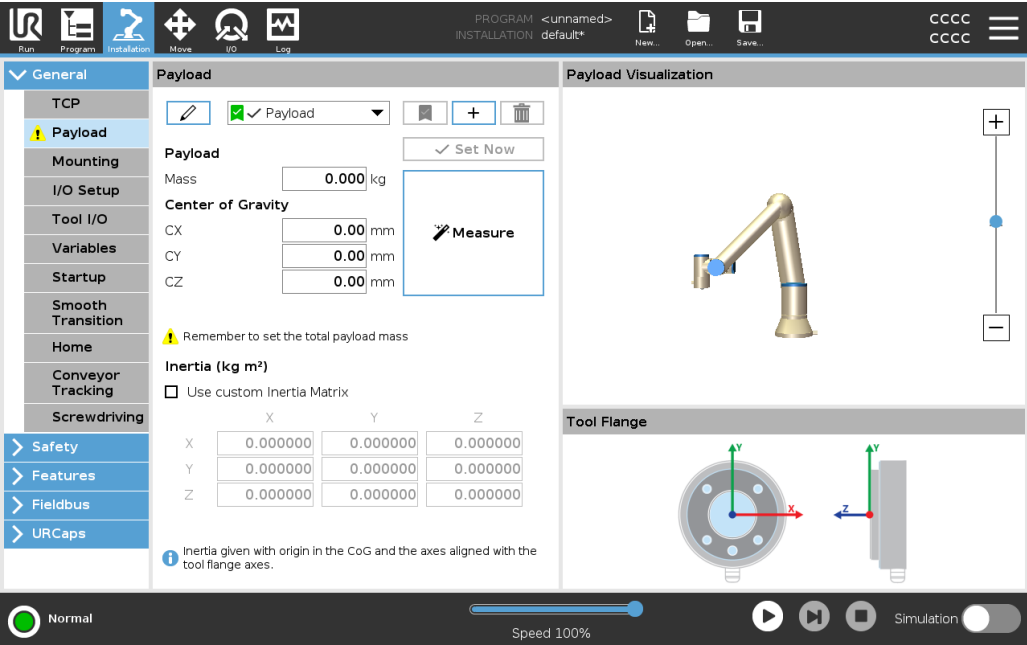
**Payload Transition Time** This is the time it takes the robot to adjust for a given payload. At the bottom of the screen, you can set the transition time between different payloads. You can add a payload transition time in seconds. Setting a transition time larger than zero, prevents the robot from doing a small "jump", when the payload changes. The program continues while the adjustment is taking place. Using the Payload Transition Time is recommended when picking up or releasing heavy objects or using a vacuum gripper.

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# 10.5.10. Payload

## Description

You must set the Payload, the CoG and the inertia for the robot to perform optimally. You can define multiple Payloads, and switch between them in your program. This is useful in Pick and Place applications, for example, where the robot picks up and releases an object.



## Adding, Renaming, Modfying and Removing Payloads

You can start configuring a new Payload with the following actions:

- Tap the to define a new Payload with a unique name. The new payload is available in the drop-down menu.
- Tap the to rename a Payload.
- Tap the to remove a selected Payload. You cannot remove the last Payload.

## Active Payload

The checkmark in the drop-down indicates which payload is active . The active Payload can be changed using the

## Default Payload

The default Payload is set as the active Payload before the program starts.

- Select the desired Payload and tap **Set as default** to set a Payload as the default.

The green icon in the drop-down menu indicates the default configured Payload



## Setting the Center of Gravity

Tap the fields CX, CY and CZ to set the center of gravity. The settings apply to the selected Payload.

## Payload Estimation

This feature allows the robot to help set the correct Payload and Center of Gravity (CoG).

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### Using the Payload Estimation Wizard

1. In the Installation Tab, under General, select **Payload**.
2. On the Payload screen, tap **Measure**.
3. In the Payload Estimation Wizard tap **Next**.
4. Follow the steps in the Payload Estimation Wizard to set the four positions. Setting the four positions requires moving the robot arm into four different positions. The load of the payload is measured at each position.
5. Once all measurements are complete, you can verify the result and tap **Finish**.



#### NOTICE

Follow the these guidelines for best Payload Estimation results:

- Ensure the TCP positions are as different as possible from each other
  - Perform the measurements within a short timespan
  - Avoid pulling on the tool and/or attached payload before and during estimation
  - Robot mounting and angle must be correctly defined in the installation
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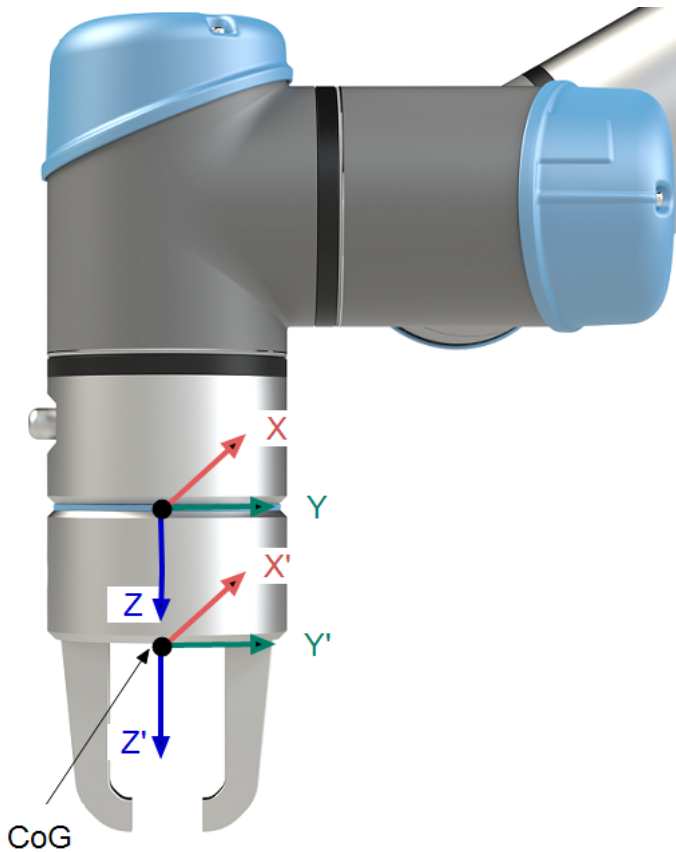
## Setting Inertia Values

You can select **Use custom Inertia Matrix** to set inertia values.

Tap the fields:  $I_{XX}$ ,  $I_{YY}$ ,  $I_{ZZ}$ ,  $I_{XY}$ ,  $I_{XZ}$  and  $I_{YZ}$  to set the inertia for the selected Payload.

The inertia is specified in a coordinate system with the origin at the Center of Gravity (CoG) of the payload and the axes aligned with the tool flange axes.

The default inertia is calculated as the inertia of a sphere with the user specified mass, and a mass density of  $1\text{g/cm}^3$



## 10.5.11. Mounting

### Description

Specifying the mounting of the Robot arm serves two purposes:

1. Making the Robot arm appear correctly on screen.
2. Telling the controller about the direction of gravity.

An advanced dynamics model gives the Robot arm smooth and precise motions, as well as allows the Robot arm to hold itself in **Freedrive Mode**. For this reason, it is important to mount the Robot arm correctly.



#### WARNING

Failure to mount the Robot's arm correctly may result in frequent robot stops, and/or the Robot arm will move when pressing the **Freedrive** button.

If the Robot arm is mounted on a flat table or floor, no change is needed on this screen. However, if the Robot arm is **ceiling mounted**, **wall mounted**, or **mounted at an angle**, this needs to be adjusted using the buttons.

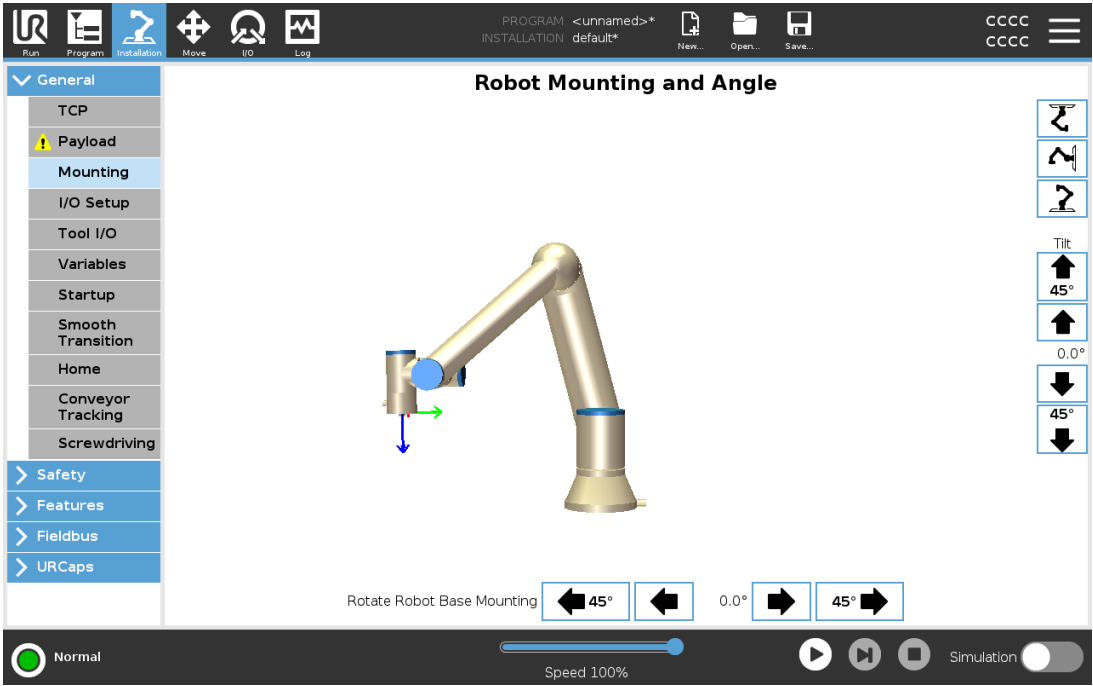
The buttons on the right side of the screen are for setting the angle of the Robot arm's mounting. The top three right side buttons set the angle to **ceiling** (180°), **wall** (90°), **floor** (0°). The **Tilt** buttons set an arbitrary angle.

The buttons on the lower part of the screen are used to rotate the mounting of the Robot arm to match the actual mounting.



**WARNING**

Use the correct installation settings. Save and load the installation files with the program.





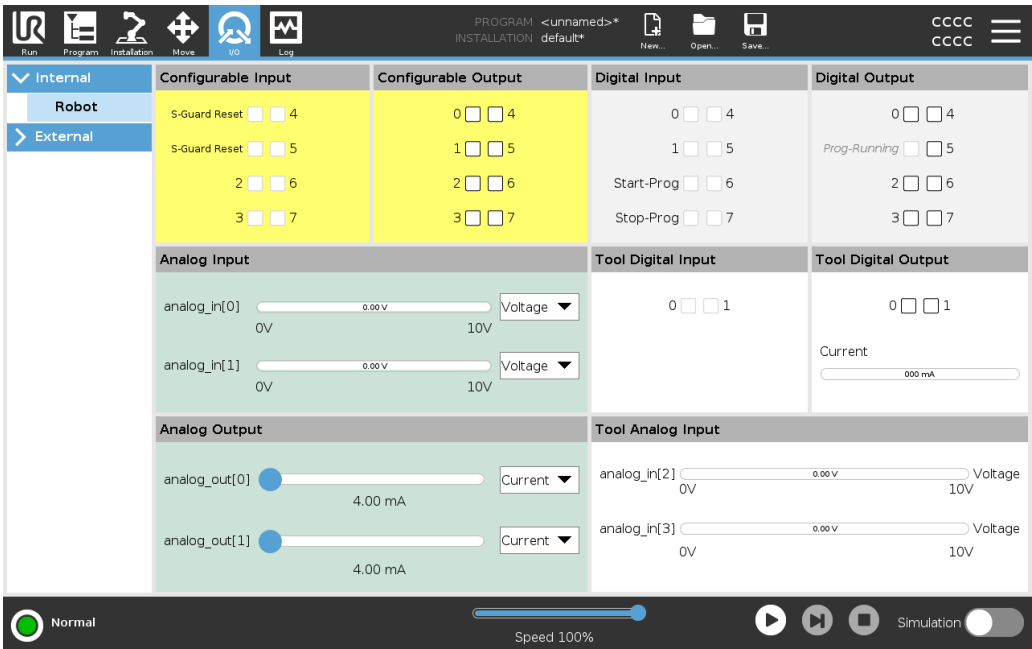
# 10.5.12. Using the I/O Tab

**Description** Use the I/O Tab screen to monitor and set the live I/O signals from/to the Control Box.

The screen displays the current state of the I/O, including during program execution. The program stops if anything is changed during execution. At program stop, all output signals retain their states. The screen updates at 10Hz, so a very fast signal might not display properly.

Configurable I/Os can be reserved for special safety settings defined in the safety I/O configuration section of the installation (see I/O); those which are reserved will have the name of the safety function in place of the default or user defined name.

Configurable outputs that are reserved for safety settings are not toggleable and will be displayed as LED's only.



**Voltage** When the Tool Output is controlled by the user, you can configure Voltage. Selecting a URCap removes access to Voltage.

**Analog Domain Settings** The analog I/O's can be set to either current [4-20mA] or voltage [0-10V] output. These settings are persistent over restarts of the robot controller and saved in the installation. Control over the tool I/Os could be assigned to a URCap in **Tool I/O** of the **Installation** tab. Selecting a URCap removes user's control over tool's analog I/O.

**Tool  
Communication  
Interface**

When the **Tool Communication Interface TCI** is enabled, the tool analog input becomes unavailable. On the I/O screen, the **Tool Input** field appears as shown.

| Tool Analog Input |        |
|-------------------|--------|
| Baud Rate         | 115200 |
| Parity            | None   |
| Stop Bits         | One    |
| RX Idle Chars     | 1.50   |
| TX Idle Chars     | 3.50   |

**Dual Pin power**

When **Dual Pin Power** is enabled, the tool digital outputs must be named as follows:

- tool\_out[0] (Power)
- tool\_out[1] (GND)

Tool Digital Output

Power

☐

☐

GND

Current

000 mA

10.5.13. Analog Input: Communication Interface

**Description**

The Tool Communication Interface (TCI) enables the robot to communicate with an attached tool via the robot tool analog input. This removes the need for external cabling. Once the Tool Communication Interface is enabled, all tool analog inputs are unavailable

**Tool  
Communication  
Interface**

1. Tap the Installation tab and under General tap Tool I/O.
2. Select Communication Interface to edit TCI settings.  
Once the TCI is enabled, the tool analog input is unavailable for the I/O Setup of the Installation and does not appear in the input list. Tool analog input is also unavailable for programs as Wait For options and expressions.
3. In the drop-down menus under Communication Interface, select required values. Any changes in values are immediately sent to the tool. If any installation values differ from what the tool is using, a warning appears.

10.5.14. Digital Output

**Description** The tool communication interface allows two digital outputs to be independently configured. In PolyScope, each pin has a drop-down menu that allows the output mode to be set. The following options are available:

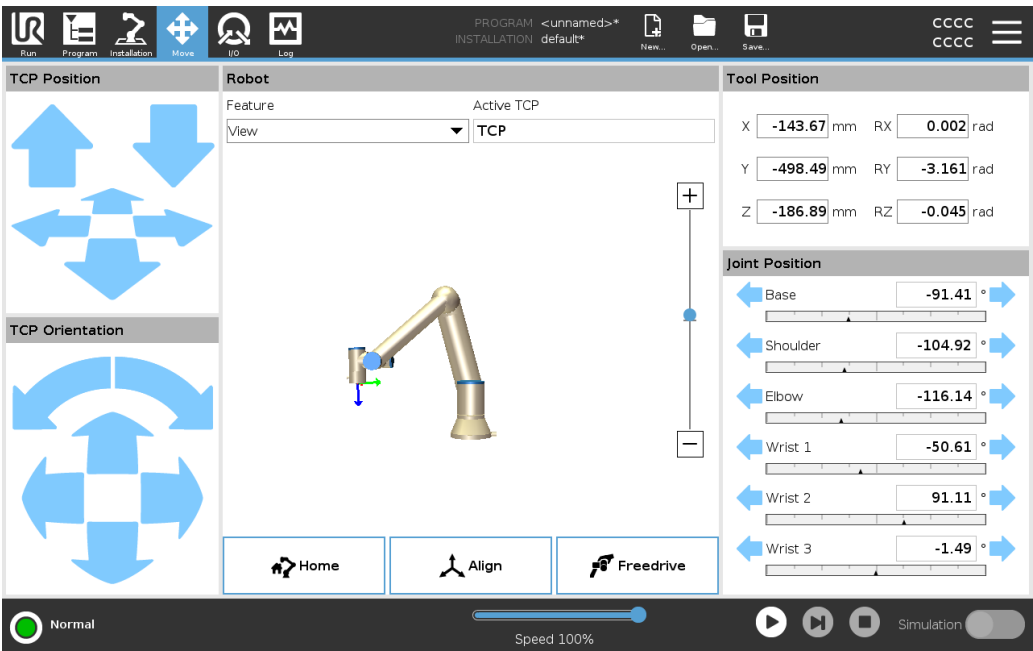
- Sinking: This allows the pin to be configured in an NPN or Sinking configuration. When the output is off, the pin allows a current to flow to the ground. This can be used in conjunction with the PWR pin to create a full circuit.
- Sourcing: This allows the pin to be configured in a PNP or Sourcing configuration. When the output is on, the pin provides a positive voltage source (configurable in the IO Tab). This can be used in conjunction with the GND pin to create a full circuit.
- Push / Pull: This allows the pin to be configured in a Push / Pull configuration. When the output is on, the pin provides a positive voltage source (configurable in IO Tab). This can be used in conjunction with the GND pin to create a full circuit. When the output is off, the pin allows a current to flow to the ground.

After selecting a new output configuration, the changes take effect. The currently loaded installation is modified to reflect the new configuration. After verifying the tool outputs are working as intended, make sure to save the installation to prevent losing changes.

**Dual Pin Power** Dual Pin Power is used as a source of power for the tool. Enabling Dual Pin Power disables default tool digital outputs.

### 10.5.15. Using the Move Tab

**Description** Use the Move Tab screen to move (jog) the robot arm directly, either by translating/rotating the robot tool, or by moving robot joints individually.



**To use the Move Tool arrows**      Hold down any of the **Move Tool** arrows to move the robot arm in the corresponding direction.

- The **Translate arrows** (upper) move the tool flange in the direction indicated.
- The **Rotate arrows** (lower) change the orientation of the tool in the indicated direction. The rotation point is the Tool Center Point (TCP), i.e. the point at the end of the robot arm that gives a characteristic point on the tool. The TCP is shown as a small blue ball.

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**Robot**      If the current position of the TCP approaches a safety plane, a trigger plane, or the orientation of robot tool is near the tool orientation boundary limit, a 3D representation of the proximate boundary limit is shown. The visualization of boundary limits is disabled during program execution.

Safety planes display in yellow and black with an arrow indicating which side of the plane, the robot TCP is allowed to be positioned.

Trigger planes display in blue and green with an arrow indicating the side of the plane, where the **Normal** mode limits are active.

The tool orientation boundary limit is visualized with a spherical cone together with a vector indicating the current orientation of the robot tool. The inside of the cone represents the allowed area for the tool orientation (vector).

When the robot TCP is no longer in proximity of the limit, the 3D representation disappears. If the TCP is in violation or very close to violating a boundary limit, the visualization of the limit turns red.

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**Feature**      Under **Feature**, you can define how to control the robot arm relative to **View**, **Base** or **Tool** features. For the best feel for controlling the robot arm you can select the **View** feature, then use **Rotate arrows** to change the viewing angle of the 3D image to match your view of the real robot arm.

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**Active TCP**      In the **Robot** field, under **Active TCP**, the name of the current active Tool Center Point (TCP) is displayed.

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**Home**      The **Home** button accesses the **Move Robot into Position** screen, where you can hold down the **Auto** button to move robot into position previously defined under Installation. The Home button's default setting returns the Robo Arm to an upright position.

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**Freedrive**      The on-screen **Freedrive** button allows the Robot Arm to be pulled into desired positions/poses.

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**Align**      The **Align** button allows the Z axis of the active TCP to align to a selected feature.

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**Tool Position** The text boxes display the full coordinate values of the TCP relative to the selected feature. You can configure several named TCPs (see ). You can also tap **Edit pose** to access the **Pose Editor** screen.

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**Joint Position** The **Joint Position** field allows you to directly control individual joints. Each joint moves along a default joint limit range from  $-360^{\circ}$  to  $+360^{\circ}$ , defined by a horizontal bar. Once the limit is reached you cannot move a joint any further. You can configure joints with a position range different from the default, this new range is indicated with red zone inside the horizontal bar.

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**Using Freedrive in the Move tab** The **Freedrive** button shall only be used in applications if allowed by the risk assessment.



**WARNING**

Failure to correctly configure the mounting setting can result in unwanted robot arm movement when you use the **Freedrive** button.

- Payload settings and robot mounting settings shall be set correctly before using Freedrive.
- All personnel shall remain outside the reach of the robot arm, when **Freedrive** is in use.



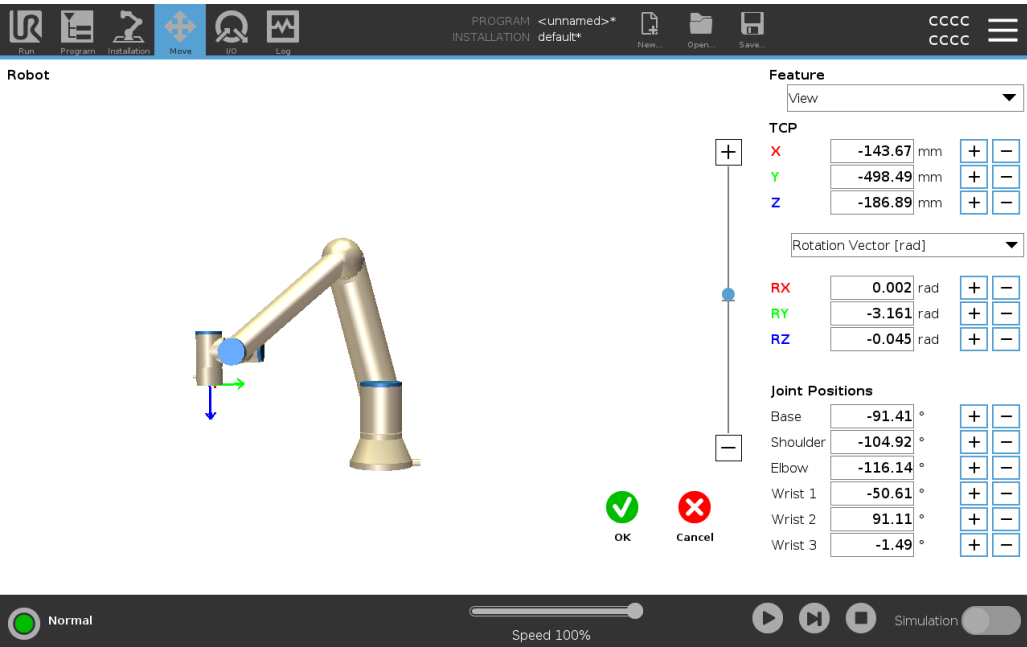
**WARNING**

Failure to correctly configure the installation settings, can increase the risk of the robot arm falling during **Freedrive**, due to payload errors.

- Verify the installation settings are correct (e.g. Robot mounting angle, payload mass and payload center of gravity offset) . Save and load the installation files along with the program.
  - Save and load the installation files along with the program.
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Description

Once you access the **Pose Editor** screen, you can precisely configure a target joint positions, or a target pose (position and orientation) for the TCP. Note: This screen is **offline** and does not control the Robot Arm directly.



**Robot**

The 3D image shows the current Robot Arm position. The **shadow** shows the Robot Arm target position controlled by the specified values on the screen. Press the magnifying glass icons to zoom in/out or drag a finger across it to change the view.

If the specified target position of the robot TCP is close to a safety or trigger plane, or the orientation of robot tool is near the tool orientation boundary limit, a 3D representation of the proximate boundary limit is shown. Safety planes are visualized in yellow and black with a small arrow representing the plane normal, which indicates the side of the plane on which the robot TCP is allowed to be positioned. Trigger planes are displayed in blue and green and a small arrow pointing to the side of the plane, where the **Normal** mode limits are active. The tool orientation boundary limit is visualized with a spherical cone together with a vector indicating the current orientation of the robot tool. The inside of the cone represents the allowed area for the tool orientation (vector). When the target robot TCP is no longer in proximity of the limit, the 3D representation disappears. If the target TCP is in violation or very close to violating a boundary limit, the visualization of the limit turns red.

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|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Feature and Tool Position</b> | <p>The active TCP and coordinate values of the selected feature are displayed. The <b>X, Y, Z</b> coordinates specify tool position. The <b>RX, RY, RZ</b> coordinates specify orientation. Use the drop down menu above the <b>RX, RY</b> and <b>RZ</b> boxes to choose the orientation representation type:</p> <ul style="list-style-type: none"><li>• <b>Rotation Vector [rad]</b> The orientation is given as a <i>rotation vector</i>. The length of the axis is the angle to be rotated in radians, and the vector itself gives the axis about which to rotate. This is the default setting.</li><li>• <b>Rotation Vector [°]</b> The orientation is given as a <i>rotation vector</i>, where the length of the vector is the angle to be rotated in degrees.</li><li>• <b>RPY [rad]</b> <i>Roll, pitch and yaw (RPY)</i> angles, where the angles are in radians. The RPY-rotation matrix (X, Y', Z" rotation) is given by:<br/><math>R_{rpy}(\gamma, \beta, \alpha) = R_Z(\alpha) \cdot R_Y(\beta) \cdot R_X(\gamma)</math></li><li>• <b>RPY [°]</b> <i>Roll, pitch and yaw (RPY)</i> angles, where angles are in degrees.</li></ul> <p>You can tap the values to edit the coordinates. You can also tap the <b>+</b> or <b>-</b> buttons to the right of a box to add/subtract an amount to/from the current value. Or you can hold down a button to directly increase/decrease the value.</p> <hr/> |
| <b>Joint Positions</b>           | <p>Individual joint positions are specified directly. Each joint position can have Joint Limit range from <math>-360^{\circ}</math> to <math>+360^{\circ}</math>. You can configure Joint Positions as follows:</p> <ul style="list-style-type: none"><li>• Tap the joint position to edit the values.</li><li>• Tap the <b>+</b> or <b>-</b> buttons to the right of a box to add or subtract an amount to/from the current value.</li><li>• Hold down a button to directly increase/decrease the value.</li></ul> <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>OK Button</b>                 | <p>If you activate this screen from the <b>Move</b> screen (see ), tap the <b>OK</b> button to return to the <b>Move</b> screen. The Robot Arm moves to the specified target. If the last specified value was a tool coordinate, the Robot Arm moves to the target position using movement type <b>MoveL</b>; or it uses movement type <b>MoveJ</b> if a joint position was specified last.</p> <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Cancel Button</b>             | <p>The <b>Cancel</b> button exits the screen discarding all changes.</p> <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |